

constrained Pareto search

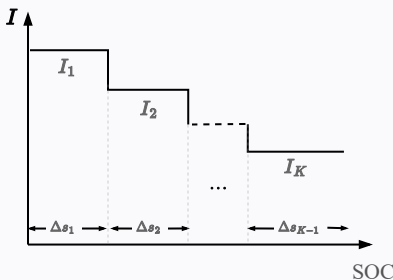
DECISION VARIABLES

$$\theta = [I_1, \dots, I_K, \Delta s_1, \dots, \Delta s_{K-1}]^T$$

K -stage multistage constant-current protocol

SOC-domain switching boundaries over interval $[s_0, s_f]$

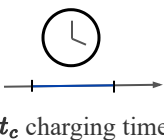
$$\theta \in \Theta_{\text{proto}} \text{ (box-feasible currents \& SOC spans)}$$



propose protocol

OBJECTIVES (ALL MINIMIZED)

$$\min_{\theta} f(\theta) = [t_c(\theta), \Delta T_p(\theta), D_{\text{chg}}(\theta)]^T$$



t_c charging time



ΔT_p peak temperature rise



D_{chg} degradation score

evaluate objectives

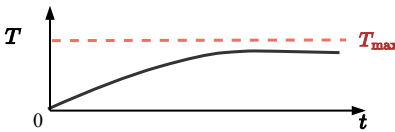
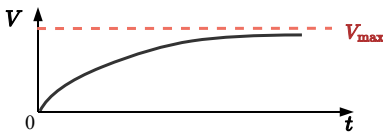
CONSTRAINTS

$$V_{\text{peak}}(\theta) \leq V_{\text{max}} \quad (\text{path voltage limit})$$

$$T_{\text{peak}}(\theta) \leq T_{\text{max}} \quad (\text{path temperature limit})$$

$$I_{\text{min}} \leq I_k \leq I_{\text{max}}, \quad \sum_k \Delta s_k = s_f - s_0, \quad \Delta s_K \geq \Delta s_K^{\text{min}}$$

(box feasibility in Θ_{proto})



Steel Blue = protocol representation

Warm Orange = objectives

Bluish Green = feasibility